2. Model-Based Intelligent Agent - Grid Navigation

Python practical code

```
grid = [
    [0, 0, 1],
    [1, 0, 0],
    [0, 0, 0]
]

start = (0, 0)
goal = (2, 2)

queue = [start]
visited = [start]

while queue:
    x, y = queue.pop(0)

    if (x, y) == goal:
        print("Goal Reached")
        break

    for dx, dy in [(0,1), (1,0), (0,-1), (-1,0)]:
        nx, ny = x + dx, y + dy

        if 0 <= nx < 3 and 0 <= ny < 3:
            if grid[nx][ny] == 0 and (nx, ny) not in visited:
                visited.append((nx, ny))
                queue.append((nx, ny))
```